WORKED SOLUTION

STEP II 1990, Question 9

Pure • Polar Curves and Arc Length

STEP II 1990, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s2-q9

The key idea

Both curves are given as r in terms of θ, so compare them ray by ray: on each half-line from the origin the parabola has exactly one point, and it is either inside or outside the cardioid. That tells us at once where the parabola enters and leaves the cardioid. For the length, the cardioid is best left in polar form (its arc-length integrand collapses with a half-angle identity), while the parabola is much easier in Cartesian form.

Solution

The two curves

For the parabola, $r + r\cos\theta = 1$ says $r = 1 - x$. Squaring, $x^2 + y^2 = (1-x)^2$, so

$$y^2 = 1 - 2x.$$

This is a parabola with focus at the origin, vertex at $(\frac{1}{2}, 0)$, opening to the left. The origin lies on its concave side, where $y^2 < 1 - 2x$; call this the inside of the parabola.

The cardioid $r = 4(1+\cos\theta)$ has its cusp at the origin (when $\theta = \pi$) and reaches $x = 8$ when $\theta = 0$.

Where they meet

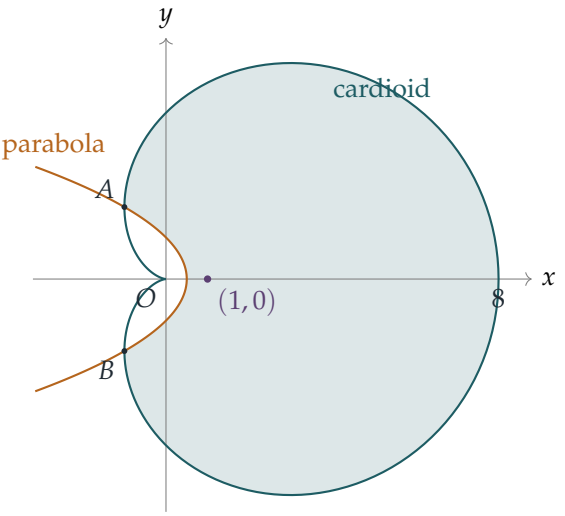
Fix a direction θ with $-\pi < \theta < \pi$, so that $1 + \cos\theta > 0$. On that ray the parabola is at distance $1/(1+\cos\theta)$ and the cardioid's boundary at distance $4(1+\cos\theta)$, and the interior of the cardioid is the segment of the ray before the boundary. So the parabola point is inside the cardioid exactly when

$$\frac{1}{1+\cos\theta} < 4(1+\cos\theta) \iff (1+\cos\theta)^2 > \tfrac{1}{4} \iff \cos\theta > -\tfrac{1}{2} \iff |\theta| < \tfrac{2\pi}{3},$$

where we multiplied by the positive quantity $1+\cos\theta$ and took the positive square root. Equality holds at $\theta = \pm\frac{2\pi}{3}$, where $r = 2$, giving the two crossing points

$$A = (-1, \sqrt{3}), \qquad B = (-1, -\sqrt{3}).$$

So the parabola is inside the cardioid along one unbroken arc, from B through the vertex $(\frac{1}{2}, 0)$ to A, and outside it everywhere else. This single arc joins two boundary points of the cardioid, so it cuts the interior into two parts: one inside the parabola (containing the origin, bounded by the small loop of the cardioid near the cusp) and one outside the parabola (bounded by the large outer arc of the cardioid).



(A neat detail visible in the sketch: $x = 4(1+\cos\theta)\cos\theta = (2\cos\theta+1)^2 - 1 \geq -1$, so A and B are also the leftmost points of the cardioid.)

The length of the boundary

The point $r = 1$, $\theta = 0$ is $(1, 0)$. It is inside the cardioid ($1 < 8$) and outside the parabola ($0^2 > 1 - 2$), so it lies in the shaded part. Its boundary consists of the cardioid arc $-\frac{2\pi}{3} \leq \theta \leq \frac{2\pi}{3}$ and the parabola arc from B to A.

Cardioid arc. In polar coordinates $\mathrm{d}s = \sqrt{r^2 + (\mathrm{d}r/\mathrm{d}\theta)^2}\,\mathrm{d}\theta$. Here $\mathrm{d}r/\mathrm{d}\theta = -4\sin\theta$, so

$$r^2 + \left(\frac{\mathrm{d}r}{\mathrm{d}\theta}\right)^2 = 16(1 + 2\cos\theta + \cos^2\theta + \sin^2\theta) = 32(1+\cos\theta) = 64\cos^2\tfrac{\theta}{2}.$$

Ask yourself Can we just write $\sqrt{64\cos^2(\theta/2)} = 8\cos(\theta/2)$?

Only because $\cos\frac{\theta}{2} \geq 0$ throughout: for $|\theta| \leq \frac{2\pi}{3}$ we have $|\frac{\theta}{2}| \leq \frac{\pi}{3}$. Using symmetry about the x-axis,

$$L_{\text{cardioid}} = 2\int_0^{2\pi/3} 8\cos\tfrac{\theta}{2}\,\mathrm{d}\theta = 32\left[\sin\tfrac{\theta}{2}\right]_0^{2\pi/3} = 32\cdot\tfrac{\sqrt{3}}{2} = 16\sqrt{3}.$$

Parabola arc. Use y as the parameter: $x = \frac{1}{2}(1 - y^2)$, so $\mathrm{d}x/\mathrm{d}y = -y$ and $\mathrm{d}s = \sqrt{1+y^2}\,\mathrm{d}y$, with y running from $-\sqrt{3}$ to $\sqrt{3}$. By symmetry,

$$L_{\text{parabola}} = 2\int_0^{\sqrt{3}} \sqrt{1+y^2}\,\mathrm{d}y.$$

With $y = \sinh u$, $\mathrm{d}y = \cosh u\,\mathrm{d}u$, the integrand becomes $\cosh^2 u = \frac{1}{2}(1 + \cosh 2u)$, so

$$\int \sqrt{1+y^2}\,\mathrm{d}y = \tfrac{1}{2}u + \tfrac{1}{4}\sinh 2u = \tfrac{1}{2}(\operatorname{arsinh} y + y\sqrt{1+y^2}) + C.$$

At $y = \sqrt{3}$, $\sqrt{1+y^2} = 2$ and $\operatorname{arsinh}\sqrt{3} = \ln(\sqrt{3}+2)$, so

$$L_{\text{parabola}} = \left[\operatorname{arsinh} y + y\sqrt{1+y^2}\right]_0^{\sqrt{3}} = \ln(2+\sqrt{3}) + 2\sqrt{3}.$$

Total. Adding,

$$L = 16\sqrt{3} + 2\sqrt{3} + \ln(2+\sqrt{3}) = 18\sqrt{3} + \ln(2+\sqrt{3}),$$

as required.

Checks

- The whole cardioid has length $\int_{-\pi}^{\pi} 8\cos\frac{\theta}{2}\,\mathrm{d}\theta = 32$, and the outer arc we used, $16\sqrt{3} \approx 27.71$, is most of it, which matches the sketch: the small loop near the cusp has length $32 - 16\sqrt{3} \approx 4.29$.
- The parabola arc, about 4.78, is a little longer than the two straight segments from B to the vertex $(\frac{1}{2}, 0)$ to A, of total length $2\sqrt{1.5^2 + 3} \approx 4.58$, as it should be.
- Numerically, midpoint-rule quadrature of the polar arc-length integrands for both curves gave a total of 32.49387, agreeing with $18\sqrt{3} + \ln(2+\sqrt{3}) \approx 32.49387$.

Common slips

- Taking the wrong part: $(1, 0)$ is outside the parabola (to the right of its vertex at $x = \frac{1}{2}$), so its boundary uses the large outer arc of the cardioid, not the loop near the cusp.
- Integrating the cardioid from 0 to 2π in one go with $8\cos\frac{\theta}{2}$, which goes negative for $\theta > \pi$; the integrand is $8|\cos\frac{\theta}{2}|$.

Beyond STEP

Both curves are conics or limaçons written with a focus or cusp at the pole, and $r = \ell/(1 + e\cos\theta)$ is the form in which conics appear in orbital mechanics, with the

Sun at the focus. The arc-length formula $\int \sqrt{r^2 + r'^2}\,\mathrm{d}\theta$ is a special case of the length of a parametrised curve, $\int |\mathbf{r}'(t)|\,\mathrm{d}t$, which in first-year vector calculus leads on to curvature and line integrals. Kleppner and Kolenkow's *An Introduction to Mechanics* uses the polar form of conics to derive Kepler's orbits.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.